

Narrative Information Theory

Lion Schulz¹ Miguel Patricio² Daan Odijk²

¹Bertelsmann ²RTL Nederland

Abstract

We propose an information-theoretic framework to measure narratives, providing a formalism to understand pivotal moments, cliffhangers, and plot twists. This approach offers creatives and AI researchers tools to analyse and benchmark human- and AI-created stories. We illustrate our method in TV shows, showing its ability to quantify narrative complexity and emotional dynamics across genres. We discuss potential applications in human-in-the-loop generative AI storytelling.

1 Background, Framework, and Results

As AI systems begin to tell their own stories [1, 2], it becomes crucial to develop formal methods for understanding and evaluating the content they produce [3, 4, 5, 6, 7]. We introduce a general information-theoretic framework to measure narratives capturing key storytelling elements like novelty and surprise [8, 9, 10]. This work provides tools for creatives and for applied machine learning scientists to analyse narrative structures and for GenAI researchers to benchmark stories told by machines. It thereby also offers a foundation for developing human-in-the-loop AI systems assisting in narrative creation while maintaining the nuanced human understanding of stories [11].

Our framework decomposes stories into states, s_t . For illustration, we applied it to a corpus of TV shows where we defined states as distributions of emotions inferred from actor faces. We show an example trajectory for an episode in Fig. 1A. Generally, our framework is applicable to other media like books [8]. We also note that our use of emotions is illustrative and our framework is agnostic towards more complex states and additional modalities.

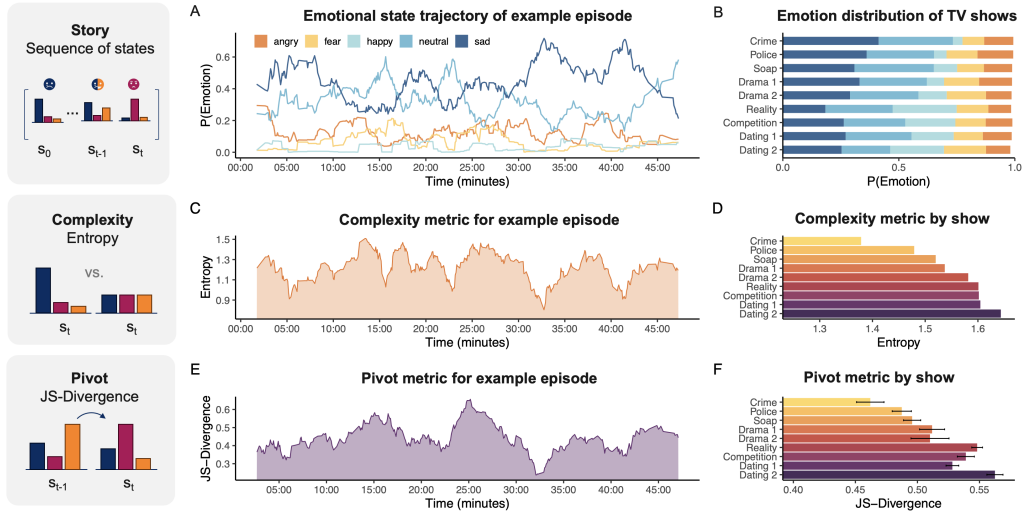


Figure 1: Information-theoretic measures of narratives.

We first introduce a measure to describe a state’s complexity [12], its entropy: **Complexity** = $\mathbf{H}(s_t)$. For our TV shows, this is lower in scenes dominated by one emotion and higher given a mixture of emotions. In our example episode, peaks of sadness are characterized by low entropy (Fig. 1C). We also apply our measure at the level of entire shows, analysing our corpus of over 3000 minutes of video (Fig. 1D and appendix). This revealed distinct patterns across genres. Reality formats showed higher entropy, indicating a broader mix of emotions, while dramas/thrillers had lower entropy, focussing on specific emotional tones. We show the underlying emotion distributions in Fig. 1B.

We can extend this approach to story dynamics, or transition between states [8, 10]. A character might start out happy and then be sad. We describe such pivots as the Jensen-Schannon-divergence between two states: **Pivot** = $\mathbf{JSD}(s_t \parallel s_{t-1})$. We show an example of this pivot measure in Fig. 1D. Our pivot measure also revealed genre-specific patterns (see Fig. 1F). Reality and dating shows displayed higher JSD, indicating frequent and stronger emotional shifts – essentially an "emotional rollercoaster" metric. In contrast, dramas and thrillers showed lower average JSD, suggesting more gradual emotional changes and slower storytelling. These results demonstrate our framework’s ability to quantify narrative structures and emotional dynamics, providing a basis for comparing storytelling techniques across different genres and between human-created and AI-generated content.

Audiences do not just observe what happens at time t but also wonder what happens next. We consider them to predict a next story state based on the story so far. We write this prediction as $P(s_{t+1}|S_t)$, essentially the output of a (generative) model m that takes $S_t = \{s_t, s_{t-1}, \dots, s_0\}$ as its input [10]. In this paper, we outline theoretically how we can understand these predictions and reserve analysis and the accompanying model development for future work (see Fig. 2 in appendix for overview). First, we can ask how predictable a story is via the mutual information between the history and a future, realised state - high mutual information means a show is predictable: **Predictability** = $\mathbf{I}(s_{t+1}; S_t)$.

A key question for an audience is how much it remains in the dark about what will happen next. We can capture this uncertainty in information-theoretic terms by computing the entropy over the predicted distribution: **Suspense** = $\mathbf{H}(P(s_{t+1}|S_t))$. If the entropy is low, a film viewer would (feel like they) clearly know what will happen in the next scene. Shows with cliffhangers would see a spike in this entropy towards the end of an episode whereas closed endings should have this entropy measure decline in the final chapters. When predictions meet reality, we can again quantify an audience’s reaction by computing the JSD between the prediction and the revealed reality of the story: **Plot twist** = $\mathbf{JSD}(P(s_{t+1}|S_t) \parallel s_{t+1})$. A high divergence will signify an unexpected plot twist. Low divergence will mean that a story treads along as expected. We note how these prediction-based measures are orthogonal to the measures introduced at the outset. Take for example our two JSD measures: changes in the state might be entirely predictable and absences of a change might be suprising if they violate predictions – imagine a character receiving good news but staying sad.

2 Discussion

Our framework provides a foundation for several key areas of interest at the intersection of creativity and AI: For example, our measures offer quantitative metrics for assessing the complexity, unpredictability, and plot twists in stories. The genre-specific patterns observed in TV shows could serve as baselines for evaluating AI-generated stories in different styles. They may also help distinguish between human and AI-generated narratives, potentially revealing characteristic patterns in AI stories, as well as help identifying systemic biases in narratives across genres or cultures – ensuring AI-generated content reflects diverse storytelling traditions. Additionally, our framework might assist human and AI co-creators, suggesting plot developments or highlighting areas that may need more tension or surprise to maintain audience engagement. For more classical ML applications, our metrics can serve as metadata for viewership analyses and recommendations. It may also help both human and AI editors identify crucial and relevant moments, accelerating summary or trailer generation [13].

Future work should focus on these applications and extend the framework to other modalities like books [8] or even music [14]. For prediction-focused measures, applying (generative) models to capture human intuitions about stories will be a key challenge. An LLM’s next token distribution might be a starting point. Creative professionals’ attitudes towards such quantitative narrative analysis tools should guide further development. In conclusion, we offer a bridge between the creative intuitions of human storytellers and the analytical capabilities of ML, contributing to the ongoing dialogue about how AI can augment and enhance human creativity rather than replace it.

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A Appendix / supplemental material

A.1 Dataset

For the analysis that is illustrated in Figure 1, we had access to an internal dataset from a large European network. This dataset encompassed over 3000 minutes of video and represents a broad overview of the landscape of the most popular TV formats. It contained:

- 16 episodes of a long-running daily soap focusing on the intertwined lives of residents of a small community (abbreviated as "Soap" in our plots).
- 12 episodes of a dating format where couples test the strength of their relationships by living with attractive singles on a tropical island ("Dating 2").
- 11 episodes of survival competition format where contestants are stranded on a remote island and must work together to overcome challenges ("Competition").
- 8 episodes of a reality series where single individuals run a bed and breakfast while searching for love ("Dating 1").
- 8 episodes of a gritty crime drama exploring the lives and conflicts within a criminal underworld ("Crime").
- 6 episodes of a thriller series that delves into the secretive world of espionage and undercover operations ("Police").

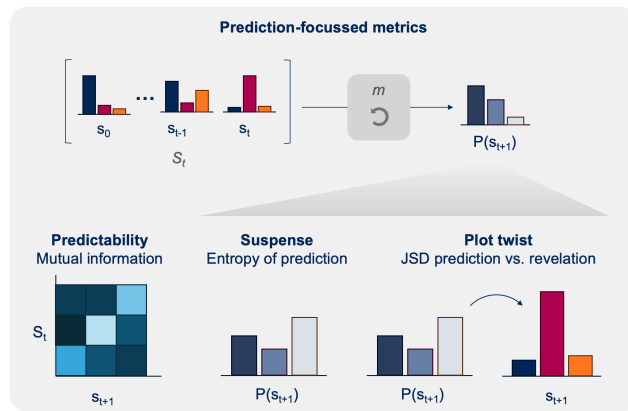


Figure 2: Supplementary figure: Visualisation of prediction-focussed metrics

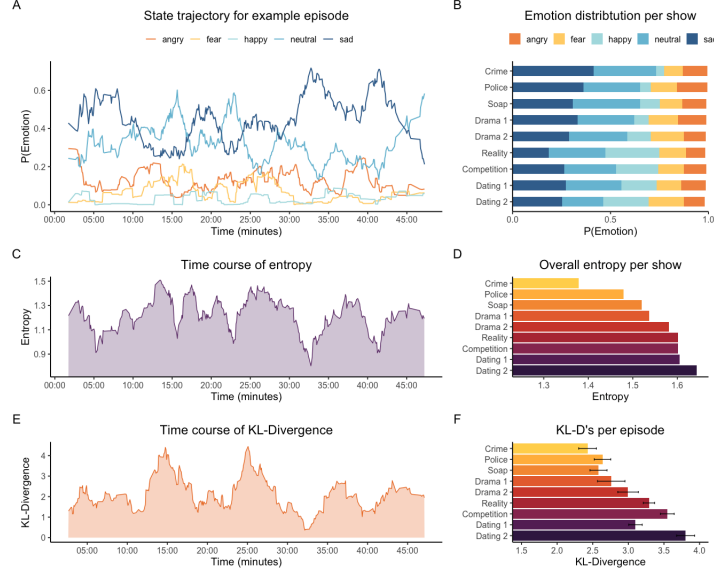


Figure 3: Supplementary figure: KL-Divergences for example show and trajectories (left) and summary statistics (right). Note the overlapping qualitative patterns with JS-D.

- 5 episodes of a crime drama based on true events, focusing on the complexities of family betrayal within a notorious crime family ("Drama 1").
- 4 episodes of a biographical drama centered on the life and challenges of a prominent public figure ("Drama 2").
- 3 episodes of a reality show following the everyday lives and humorous antics of a well-known family ("Reality").

A.2 Analysis pipeline

To analyse the individual episodes, we applied a machine learning pipeline. In short, this pipeline adhered to the following steps. We first subsampled the videos to extract a frame every 5 seconds. On these frames, we then applied face detection. On the detected faces, we applied emotion detection using the deepface library. This library outputs a distribution over 7 emotions ("angry", "disgust", "fear", "happy", "sad", "surprise", "neutral").

For our analysis, we then applied the following steps: We first averaged the distributions per frame (here using a rolling average over 20 extracted frames that contain a face). Fig. 1A plots these averaged values for the distributions of an episode of the "crime" series. We plot the entropy of these averaged states in Fig. 1C. Fig. 1D in turn shows the JSD between raw states, again averaged over 20 frames with faces. We note that we only plot the most prominent 5 emotions in Fig. 1A and B for readability - which make up more than 95 % of the emotions in the shows. Entropies and JSD in turn were computed on the seven original emotions.

To compute entropies per show, we averaged all states across episodes to form an average emotion distribution per show (Fig. 1B), and computed the entropy based on this single distribution per show (Fig. 1D). We used this single distribution as the basis because it allowed us the best high-level overview of the emotions contained in a show. In turn, to compute average values for the pivot-metric, we first averaged the JSD per episode, and then plotted the per-show means of these episode-level JSD's and the accompanying standard error of the mean in Fig. 1F.

A.3 Choice of divergence measure

We here choose JSD over Kullback-Leibler Divergence because of its symmetry and because its boundedness makes it more suitable for usage as a metric [15]. We observe similar qualitative results with KLD, see Fig. 3.